

Value of Information

Lecture

Dr. James Doss-Gollin

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1 The Umbrella Problem

Consider the simplest possible decision under uncertainty. You're heading out the door and must decide whether to bring an umbrella:

- Bringing the umbrella costs you some hassle: C
- If it rains and you don't have one, you get soaked: loss L
- The probability of rain is p

The expected cost of bringing the umbrella is simply C (you pay regardless of weather). The expected cost of leaving it is pL (you pay the loss only when it rains, which happens with probability p).

The optimal rule is: **bring the umbrella** whenever $C < pL$, i.e., when

$$p > \frac{C}{L} \quad (1)$$

The ratio C/L is the **cost-loss ratio**. It's the probability threshold above which protection is worthwhile. If protection is cheap relative to the potential loss, you should protect even at low probabilities.

To make this concrete, suppose $C = \$5$ and $L = \$50$, so $C/L = 0.1$. The payoff table is:

	Bring umbrella	Don't bring
Rain (p)	-5	-50
No rain ($1 - p$)	-5	0

For what values of p would you leave the umbrella at home? From Equation 1, you need $p < C/L = 5/50 = 0.10$. Only if the chance of rain is below 10% is it worth skipping the umbrella.

Now suppose a weather oracle offers to tell you **for certain** whether it will rain. With perfect information, you bring the umbrella *only when it rains* — you protect when needed and avoid the hassle when it's dry.

- Without the oracle: you either always protect (C) or always gamble (pL), whichever is cheaper
- With the oracle: you pay C only a fraction p of the time

How much should you pay the oracle? That question — the maximum you’d pay for perfect information before making your decision — is the **expected value of perfect information** (EVPI).

2 Expected Value of Perfect Information

The core idea of VOI is simple: **compare your optimized decision with the information to your optimized decision without it**. The difference is what the information is worth.

For EVPI, this means comparing two ways to make a decision. Let a denote the action (e.g., dike height), s the uncertain scenario, $p(s)$ the probability of scenario s , and $U(a, s)$ the payoff from taking action a in scenario s .

Without information (“average then optimize”), you must choose the best action *before* learning which scenario is true. You evaluate each action by averaging its payoff over all scenarios, then pick the best one:

$$\max_a \int U(a, s) p(s) ds \quad (2)$$

With perfect information (“optimize then average”), you choose the best action *after* learning which scenario is true. For each scenario, you pick the action that maximizes payoff in that scenario. Since you don’t know in advance which scenario you’ll observe, you take the expectation:

$$\int \max_a U(a, s) p(s) ds \quad (3)$$

The EVPI is the gap between Equation 3 and Equation 2:

$$\text{EVPI} = \int \max_a U(a, s) p(s) ds - \max_a \int U(a, s) p(s) ds \quad (4)$$

This is always ≥ 0 . The reason is Jensen’s inequality: the max function is convex, so $E[\max] \geq \max[E]$. Intuitively, information can never make you worse off — you can always ignore it and revert to your prior decision.

The EVPI has a clean interpretation: it is the **maximum you should pay a clairvoyant**. If the EVPI is small, don’t bother gathering more information. If it’s large, invest in learning.

2.1 Worked Example

Return to the umbrella with $p = 0.30$, $C = 5$, $L = 50$. Here the scenarios are discrete (rain or no rain), so the integrals in Equation 4 become sums.

Without information (Equation 2): bring the umbrella (cost = 5), since $5 < 0.3 \times 50 = 15$.

With perfect information (Equation 3):

- When it rains (prob 0.3): bring umbrella, cost = 5
- When it doesn’t rain (prob 0.7): leave it, cost = 0
- Expected cost = $0.3 \times 5 + 0.7 \times 0 = 1.50$

$$\text{EVPI} = 5.00 - 1.50 = \$3.50$$

You'd pay up to \$3.50 for a perfect rain forecast. The savings come entirely from avoiding unnecessary protection on dry days.

2.2 When is EVPI Zero?

$EVPI = 0$ when the **same action is optimal under all scenarios**. In that case, learning which scenario is true wouldn't change your decision.

Example: if $p = 0.50$ and $C/L = 0.1$, you bring the umbrella regardless — a forecast wouldn't change anything.

This is an important conceptual point: **uncertainty and information value are not the same thing**. You can have enormous uncertainty about which scenario will occur but zero EVPI, if the uncertainty doesn't affect which action is best. Conversely, a small amount of uncertainty can have high EVPI if you're right on the decision boundary.

3 The Cost-Loss Ratio

A. H. Murphy, R. W. Katz, R. L. Winkler, and W.-R. Hsu [1] generalize the umbrella problem into a canonical framework for valuing weather forecasts. The setup is the same:

- **Protect** (cost C): no loss regardless of weather
- **Don't protect**: lose L if the adverse event occurs

The innovation is distinguishing three types of information available to the decision-maker:

1. **Climatological information**: you only know the base rate p_θ (the long-run frequency of the adverse event)
2. **Perfect information**: you know exactly whether the event will occur
3. **Imperfect forecasts**: a forecast system with hit rate $p_1 = P(\text{event} \mid \text{forecast yes})$ and miss rate $p_0 = P(\text{event} \mid \text{forecast no})$

With only climatological information, the minimum expected expense is:

$$E_C = \min(p_\theta L, C) \quad (5)$$

You either always protect or never protect, whichever is cheaper. With perfect forecasts, you protect only when the event occurs:

$$E_P = p_\theta C \quad (6)$$

The value of perfect forecasts is:

$$V_P = E_C - E_P \quad (7)$$

This is our EVPI again, expressed in Murphy's notation. It is positive whenever $0 < p_\theta < 1$ and $C < L$.

Real forecasts aren't perfect. An imperfect forecast is characterized by its conditional probabilities:

- $p_1 = P(\text{event} \mid \text{forecast of event})$ — the hit rate
- $p_0 = P(\text{event} \mid \text{forecast of no event})$ — the miss rate

A key result, due to Blackwell (1953), is that the value of any properly calibrated forecast is bounded:

$$0 \leq V_F \leq V_P \quad (8)$$

Imperfect forecasts can **never have negative expected value** — as long as they’re properly calibrated, acting on them is always at least as good as ignoring them. But the relationship between forecast quality and forecast value is **nonlinear**:

- When $C/L \approx p_\theta$ (you’re “on the fence” about whether to protect), even a modest forecast improvement has high value
- When C/L is far from p_θ (the decision is obvious either way), even a large improvement in forecast quality has little value

The practical implication: forecasts are most valuable to decision-makers who are near the decision boundary. If the right action is obvious, no amount of forecast improvement will change anything.

4 Partial Information: Which Uncertainty to Resolve?

Real decisions involve **many** sources of uncertainty. For flood risk, these include the SLR trajectory, the discount rate, the surge distribution, the depth-damage function, and more. Resolving all of them is expensive or impossible. But not all of them affect the decision.

If you can only learn about *one* source of uncertainty, which one should it be?

4.1 An Example: Flood Insurance Pricing

Suppose you’re an insurer pricing flood policies. Your premium depends on uncertain inputs: the local flood probability, the building’s elevation, the depth-damage relationship, and so on. Which uncertainty should you invest in reducing?

If you’re insuring **100,000 houses in the same floodplain**, investing in a better flood model pays off handsomely. The flood hazard uncertainty is shared across every policy in the portfolio — resolving it improves all 100,000 pricing decisions at once.

But if you’re insuring **one house**, improving the regional flood model barely changes your premium. You’d be better off sending an inspector to measure the building’s actual first-floor elevation and check for flood-proofing — that building-specific uncertainty is what drives your pricing error for this particular policy.

The “most uncertain” input isn’t necessarily the one worth resolving. What matters is which uncertainty, if resolved, would most change your decision. That’s what EVPXI measures.

4.2 The EVPXI Formula

The same principle applies formally. We decompose the uncertain scenario s into two parts: s_i (one source of uncertainty, e.g., the SLR trajectory) and s_i^c (everything else, e.g., the discount rate and surge parameters).

If we learn s_i but everything else stays uncertain, we can pick the best action for each value of s_i , while still hedging over s_i^c . For a given revealed value of s_i , the best we can do is:

$$\max_a \int U(a, s_i, s_i^c) p(s_i^c | s_i) ds_i^c \quad (9)$$

This is the optimal decision conditional on knowing s_i , averaging over the remaining uncertainty s_i^c .

Since we don't know in advance which s_i we'll observe, we average Equation 9 over s_i :

$$\int \left[\max_a \int U(a, s_i, s_i^c) p(s_i^c | s_i) ds_i^c \right] p(s_i) ds_i \quad (10)$$

The **expected value of partial perfect information** (EVPXI) is the improvement of Equation 10 over the prior decision (Equation 2):

$$\text{EVPXI}(s_i) = \int \left[\max_a \int U(a, s_i, s_i^c) p(s_i^c | s_i) ds_i^c \right] p(s_i) ds_i - \max_a \int U(a, s) p(s) ds \quad (11)$$

Reading Equation 11 from left to right: resolve s_i , optimize given what you learned (but still average over everything you don't know), then average over what you might learn. Subtract the best you could do without any information.

By comparing EVPXI across different uncertainty sources, you can **prioritize research and monitoring investments**. Spend money learning about the uncertainties with the highest EVPXI, not necessarily the “most uncertain” parameters.

An important caveat: partial EVPIs are **not additive**.

$$\text{EVPXI}(A) + \text{EVPXI}(B) \neq \text{EVPXI}(A \cup B) \quad \text{in general} \quad (12)$$

The information from A may change which action is best, and that changes the value of learning B . Uncertainties can be **complementary** (together worth more than the sum) or **substitutes** (together worth less).

5 VOI Requires Probabilities

Notice that **every VOI calculation requires a probability distribution $p(s)$ over the scenarios**. The integrals in Equation 4 and Equation 11 are only defined if you can write down $p(s)$. Without probabilities, these quantities are undefined.

This is a real limitation. In many climate risk problems, we face **deep uncertainty**: situations where we cannot confidently assign probabilities to the scenarios. We might disagree about the right probability model, or the range of plausible futures might be so wide that any single distribution feels arbitrary.

When that happens, VOI breaks down — and we need different tools. That is exactly what we'll tackle in Week 8: robustness and decision-making under deep uncertainty, where the goal shifts from “optimize given probabilities” to “find strategies that perform acceptably across many possible futures.”

6 Looking Ahead

Wednesday: Global Sensitivity Analysis — a complementary tool. VOI asks “which uncertainty affects the *decision*?” Sensitivity analysis asks “which uncertainty affects the *output*?” These are not the same thing, and understanding both is essential.

Friday (Lab 6): You’ll compute EVPI and EVPXI on the ICOW flood risk problem from Lab 5, determining whether the SLR trajectory or the discount rate matters more for the dike height decision.

6.1 References

Bibliography

- [1] A. H. Murphy, R. W. Katz, R. L. Winkler, and W.-R. Hsu, “Repetitive Decision Making and the Value of Forecasts in the Cost-Loss Ratio Situation: A Dynamic Model,” *Monthly Weather Review*, vol. 113, no. 5, pp. 801–813, May 1985, doi: 10.1175/1520-0493(1985)113<0801:RD-MATV>2.0.CO;2.